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Alpha Report

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Introduction

This report presents the results of a predictive model developed to simulate the 2026 NCAA men's basketball tournament. A logistic regression model was trained using historical game data and four engineered performance metrics derived from the "Four Factors" of basketball analytics: effective field goal percentage, turnover rate, offensive rebounding rate, and free throw rate. This model achieved an AUC greater than 0.70, indicating strong predictive performance in distinguishing winners from losers.

Each feature represents the difference in performance between two teams, allowing the model to evaluate relative strengths in a matchup setting. Using these features, the model generates win probabilities for any pair of teams and was applied to simulate the full tournament bracket.

The objective of this analysis is to identify three key outcomes: the most likely national champion, a potential Cinderella team that is undervalued by its seed, and a highly seeded team that is most vulnerable to an early upset. These insights demonstrate how statistical modeling can reveal strengths and weaknesses that may not be fully captured by tournament seeding alone.

Model Prediction: National Champion

Based on the logistic regression model and simulated tournament results, Arizona is predicted to be the most likely national champion. Arizona advanced through each round by consistently outperforming its opponents across multiple statistical categories, particularly in effective field goal percentage, which is the strongest predictor identified in earlier analysis.

In the championship matchup against Florida, Arizona was given an approximate win probability of 52.7%, indicating a competitive but favorable edge. While the margin is not overwhelming, the model consistently favored Arizona due to its balanced performance across all four factors, rather than reliance on a single dominant metric.

Arizona's path to the title reflects a team that maintains efficiency, limits mistakes, and performs well in key areas such as shooting and rebounding. These characteristics align closely with the statistical profile of past championship teams, reinforcing the model's selection. Overall, Arizona emerges as the most complete team in the simulated tournament.

Cinderella Team (Most Undervalued)

The model identifies High Point (12 seed) as the most undervalued team in the tournament and the strongest candidate for a Cinderella run. Despite its lower seed, High Point advanced to the Sweet 16 by defeating higher-seeded teams such as Wisconsin and Arkansas.

This success is explained by High Point's strong differential metrics, particularly in shooting efficiency and rebounding. The model consistently favored High Point in its matchups, suggesting that its statistical profile more closely resembles that of a higher-seeded team.

The discrepancy between seeding and statistical strength indicates that High Point may be under-seeded relative to its true ability. This makes it a classic Cinderella team, where advanced metrics reveal hidden potential that is not immediately apparent from rankings alone.

High Point's performance highlights the value of data-driven analysis in identifying overlooked teams capable of outperforming expectations in a single-elimination tournament.

Fade Team (Most Likely Early Upset)

The model identifies Kansas (4 seed) as the top-seeded team most likely to be eliminated early. In the simulation, Kansas was predicted to lose in the first round to Cal Baptist, a significantly lower-seeded opponent.

This outcome is driven by Kansas's weaker differential metrics in key areas such as turnover rate and rebounding. While Kansas may appear strong based on seeding alone, its statistical profile suggests vulnerabilities.

The model's prediction demonstrates that high seeds are not always indicative of strong underlying performance. In this case, Kansas does not exhibit the same level of dominance across the Four Factors as other top teams, making it more susceptible to an upset.

This result reinforces the idea that tournament outcomes are influenced not only by rankings, but also by matchup-specific statistical advantages, which can favor lower-seeded teams.

Summary of Findings

This analysis demonstrates the effectiveness of using statistical modeling to predict outcomes in the NCAA tournament. By leveraging differential performance metrics based on the Four Factors, the logistic regression model was able to simulate the tournament and generate meaningful insights.

Arizona was identified as the most likely national champion due to its consistent and balanced performance across all key metrics. High Point emerged as the primary Cinderella team, outperforming expectations based on strong underlying statistics. Kansas was highlighted as the most vulnerable top seed, showing that traditional rankings do not always reflect true team strength.

Overall, the results emphasize that shooting efficiency remains the most influential factor in determining game outcomes, while turnover control, rebounding, and free throw rate provide additional context. The model highlights how data-driven approaches can uncover hidden strengths and weaknesses. These results demonstrate the practical value of predictive modeling in evaluating tournament outcomes beyond traditional rankings.